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31 July 1963
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MILITARY SPECIFICATION

GYPSUM CEMENT

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one grade of gypsum cement, for use in the manufacture and assembly of level vials of fire control equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves; Standard, For Testing Purposes.

PPP-C-96 - Cans, Metal, 28 Gauge and Lighter

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 5610

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MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

Method C-266-58 - Test for Time of Setting of Hydraulic Cement by Gillmore Needles.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material.- The material shall be a high-quality calcined gypsum meeting the requirements hereinafter specified.

3.2 Fineness.- All the material shall pass a No. 20 U.S. Standard test sieve, while not less than 95 percent shall pass a No. 100 sieve, when tested as specified in 4.5.1.

3.3 Setting time.- The gypsum cement shall set in not less than 25 minutes nor more than 35 minutes from the initial time of addition of water to the material. The setting time shall be determined as specified in 4.5.2.

3.4 Shrinkage and expansion.- The gypsum cement shall not shrink more than 0.10 percent, when tested as specified in 4.5.3. The cement shall not expand when so tested.

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3.5 Compression strength.- The average compression strength of the dry set gypsum cement shall be not less than 3,400 psi, when tested as specified in 4.5.4. The compression strength of any one of the individual test specimens shall be not less than 2,890 psi.

3.6 Workmanship.- The gypsum cement shall be a dry, powdered material, smooth in texture, and free of lumps and extraneous matter. The material shall be processed in accordance with high-grade commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perfrom any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.- A lot of gypsum cement shall be of uniform composition, produced under like containers in one unchanged process by one manufacturer and presented for acceptance at one time.

4.3 Sampling

4.3.1 Sampling for examination.- A random sample of shipping containers shall be selected for examination of the cement and for the preparation for delivery in accordance with level 1 of Standard MIL-STD-105.

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4.3.2 Sampling for tests.- Five containers shall be taken at random from the lot. A composite 2-pound sample shall be prepared by removing equal portions of material from each selected container, and mixing them thoroughly. The composite sample shall be placed in a tightly sealed metal container, labeled to show the name of the material, lot number, lot size, contract or order number, and name of the contractor. All acceptance tests shall be performed on material drawn from the composite sample.

4.4 Examination

4.4.1 Visual.- Sample units selected in accordance with 4.3.1 shall be examined for the visual defects in the gypsum cement and at the acceptable quality level shown in table I.

4.4.2 Packaging, packing and marking.- Sample units selected in accordance with 4.3.1 shall be examined for the defects and at the acceptable quality level shown in table II.

Table I - Examination of gypsum cement

Material	AQL percent	Classification of defects	Defects	Method of examination
Gypsum cement (See 3.6 and 4.4.1)	2.5	Major 101	Not dry	Visual and tactile
		Major 102	Not powdered	Visual and tactile
		Major 103	Not smooth in texture	Visual and tactile
		Major 104	Not free from lumps	Visual
		Major 105	Contains extraneous matter	Visual

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Table II. - Examination of packaging and packing

Item	AQL percent	Classification of defects	Defect	Method of inspection
Level A				
		Critical	None defined	
Unit container (see 4.4.2, section 5)	4.0	Major 106 Major 107 Major 108 Major 109	Improper types Improper sizes Improper fill 1/ Improper closure	Visual Visual Approved scale 2/ Visual
Box open (see 4.4.2 and section 5)	4.0	Major 110 Major 111	Improper type of box Lack of or improper strapping	Visual Visual
Box closed (see 4.4.2 and section 5)	4.0	Major 112 Major 113 Major 114	Gross weight, max. Improper marking Improperly closed	Approved scale 2/ Visual Visual
Level C				
		Critical	None defined	
Filled sacks (see 4.4.2 and section 5)	4.0	Major 115 Major 116 Major 117	Improper weight Marking, misleading or unidentifiable Improperly closed	Approved scale 2/ Visual Visual

1/ The actual weight of a container filled with the minimum required quantity of cement shall be the basis for determining the acceptable weight of subsequent containers.

2/ Approved by procuring activity

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4.5 Tests

4.5.1 Fineness.- Place a No. 20 and a No. 100 sieve, conforming to Specification RR-S-366, over a collecting pan. Place approximately 100 grams of the material, accurately weighed, on the top sieve and shake the assembly for 30 minutes in a Ro-Tap machine, or equivalent, geared to impart 300 ± 15 gyrations and 150 ± 1 taps per minute. Weigh the material retained on each sieve, and calculate the percentage of material passing each sieve.

4.5.2 Setting time.- Place a 100-gram portion of the cement in a suitable mixing container, and pour exactly 41 ml. of distilled water held at room temperature (77 ± 5°F), over the material. Record the initial time of addition of the water. Stir the mixture for 2 minutes to form a smooth paste. Immediately pour the paste into a round dish measuring approximately 2-1/2 inches in diameter and 1/2 inch in depth (see 6.2). The mixture shall be tested for setting 25 minutes after the initial addition of the water, and again 10 minutes later, using Gillmore needles conforming to weight, diameter, and form to the "final needles" described in ASTM C-26658. The needle plunger shall be released at the specified intervals of 25 and 35 minutes, and quickly reengaged 2 ± 1/5 seconds after release in each instance. The needle shall penetrate to the bottom of the dish at the 25-minute interval, to indicate non-setting of the cement, while the needle penetration shall not exceed 0.5 mm. at the 35-minute interval, to indicate that the material has set.

4.5.3 Shrinkage and expansion.- Place a 300-gram portion of the material in a suitable mixing container, and pour exactly 1,234 ml. of distilled water held at room temperature (77 ± 5°F), over the material. Record the initial time of addition of the water. Allow the water to soak into the material for 2 minutes, and then stir the mixture for 2 additional minutes to form a smooth paste.

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Immediately pour the paste into a 1 by 1 by 9-inch mold, and allow it to set for at least 1 hour. Remove the block from the mold and taper-trim both ends of the block to form 1/4 by 1/4-inch ends. The surface of the trimmed ends shall be made smooth and flat. Measure the length of the trimmed block to the nearest 0.001-inch by means of a vernier caliper. Place the block in an atmosphere conditioned to a temperature of 70 to 80°F, and a relative humidity of 30 to 50 percent. When the block has dried completely, remeasure its length to the same degree of accuracy as before. The block shall be considered completely dried when 2 consecutive weighings, made at 24-hour intervals, agree within 0.01 percent of each other.

No increase in the initial length of the test block shall occur (see 3.4). Calculate the percentage of shrinkage as follows:

$$\text{Percent shrinkage} = \frac{(\text{Initial length} - \text{dried length})}{\text{Initial length}} \times 100$$

4.5.4 Compression strength

4.5.4.1 Specimens.- Prepare a paste of the material with distilled water, mixing the ingredients in the proportion of 41 ml. of water to every 100 grams of the gypsum cement used, as specified in 4.5.2. With the prepared paste fill 5 metal cubic molds.

4.5.4.2 Molds.- Molds for making test specimens shall be 2-in. split cube molds made of noncorrodible material and sufficiently rigid to prevent spreading during molding. The mold shall have not more than 3 cube compartments and be separable into not more than 2 parts. When assembled, the parts of the molds shall be firmly held together, and dimensions shall conform to the following requirements: Interior faces shall be plane surfaces with a maximum variation of 0.001-in. for new molds and 0.002-in. for old molds: distance between opposite faces, and height of the molds, measured separately for each cube compartment

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shall be 2 ± 0.005-in. for new molds or 2 ± 0.020 in. for old molds; angle between adjacent interior faces and between interior faces and top and bottom planes of the mold shall be 90 ± 0.05 deg. measured at points slightly removed from the intersection of the faces.

4.5.4.3 Procedure.- Allow the material in the molds to set for 1 hour. Remove the set blocks from the molds, and dry them completely in an atomosphere conditioned to a temperature of 70 to 80°F and a relative humidity of 30 to 50 percent, as specified in 4.5.3. Mark the top side of the blocks so that each block may be placed in the compression testing machine with its marked side up. Place a block in the machine and apply the compression load at the rate of 20,000 ± 2,000 pounds per minute, until the block is crushed. Record the load required to crush each of the 5 blocks. Calculate the average compression strength, in terms of pounds per square inch. The materials shall be rejected if the compression strengths are not as indicated in 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.- Gypsum cement shall be packaged 1-pound net weight in metal containers of suitable size conforming to type V, class 2 of Specification PPP-C-96.

5.1.2 Level C.- Unless otherwise specified, gypsum cement shall be furnished in quantities as specified (see 6.1) in accordance with the manufacturer's commercial practice.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A.- Containers as specified in 5.1.1 shall be packed in accordance with the overseas shipment requirements of the appendix to Specification PPP-C-96.

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5.2.2 Level B.- Containers as specified in 5.1.1 shall be packed in accordance with the domestic shipment requirements of the appendix to Specification PPP-C-96.

5.2.3 Level C.- Packing shall be in accordance with commercial practice adequate to ensure acceptance and safe delivery by the carrier for the mode of transportation employed.

5.3 Marking.- In addition to any special marking specified by the procuring activity (see 6.1) all containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(c) If level C Weight of bag required (see 5.1.2)

(d) Type of paper sack if other than that of commercial practice (see 5.1.2 and 6.3).

(e) Special marking if required (see 5.3)

6.2 A dish (such as Catalog No. 4357, A. H. Thomas Co., Philadelphia, Pa.) has been found to be a suitable container for holding the cement paste in making the setting time determination (see 4.5.2).

6.3 Gypsum cement conforming to this specification has been packed in 100-pound shipping paper sacks conforming to type II, IV or V as specified, of construction number 12X of Specification UU-S-48 for immediate use with no further packing.

Custodians:
Army-MR
Navy - Docks
Air Force - MOA

Preparing activity:
Army-MR
(Project No. 5610-0029)

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